FlavorCraft Labs

Research & Planning to SEO & Analytics Documentation

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Customer Research

Finding	Evidence	Possible implication
Ingredients are critical	4/4 consider ingredients when choosing a recipe; several want them immediately	Make ingredients easy to find on recipe details
Time matters contextually	4/4 mention preparation/cooking time or quick meals; quick meals are particularly relevant during weekdays	Make preparation/cooking time visible;consider time-based discovery/filtering
Dietary filtering has varying importance	Some participants have specific dietary/filter needs; others consider filtering useful but less important	Provide clear dietary/category filtering without making it the sole focus of discovery
Saving/planning has value	3/4 responded positively to favorites, planning or shopping-list functionality	Favorites are supported by the brief; broader planning features could remain future scope
Clear instructions are important	Participants mention confusing steps, vague measurements and insufficient guidance as frustrations	Keep instructions clear, structured and easy to follow; visual guidance may also help
Users want direct, uncluttered access to	Participants mention long scrolling, ads, cluttered homepages and difficult-to-find search/categories	Prioritize recipe content and discovery controls; minimize

recipe		unnecessary content and navigation friction
Images influence recipe selection	Several participants explicitly choose recipes partly based on appealing photos; others want imagery immediately	Give recipe imagery strong visual prominence, particularly during recipe discovery

Research note:

These findings are based on four qualitative interviews and should be treated as patterns within this participant group rather conclusions about all users.

Research → SEO & Analytics

The following table translates customer research findings into technical SEO considerations and analytics questions. These questions informed the measurement strategy and usability testing; they do not represent confirmed behavioral findings.

Research Finding	SEO Implication	Analytics question(s)
Ingredients are critical	Recipe ingredients should be accurately represented in Recipe JSON-LD so search engines can understand the ingredients associated with each recipe	● Do users reach the ingredients section, or leave the recipe before reaching it? ● How far do users typically scroll before reaching the ingredients? ● Do users pause/engage around the ingredients section ● After viewing ingredients, do users continue to the instructions or leave the recipe?

		• Is viewing the ingredients associated with further actions, such as saving the recipe?
Time matters contextually	Accurately represent recipe preparation, cooking and total time in Recipe structured data and visible recipe content. Where supported by the recipe data, relevant descriptors such as “quick” or “easy” may help communicate recipe suitability for time-sensitive searches.	• How does time information influence recipe discovery and selection? • Do users interact with time-related filters/tags if available, and is there different engagement with recipes identified as quick?
Dietary filtering has varying importance	Accurately expose relevant dietary tags in the recipe content and structured data so search engines can better understand which recipes are suitable for searches such as “vegan” or “glutenfree” recipes.	• Which filters do users interact with most and least? • How do users combine filters during recipe discovery? • Are dietary filters being used, and do users

		continue to recipe results after applying them?
Saving/planning has value	No direct SEO implication identified	● Do users use the Favorites feature? ● How far through the recipe content do users typically navigate before saving a recipe, and what content have they reached before doing so?
Clear instructions are important	Accurately represent the recipe's individual instruction steps in Recipe JSON-LD so search engines understand the recipe method and its sequence.	● How do users navigate through the recipe instructions? ● Are there repeated scrolling/navigation patterns between instructions, ingredients and other recipe content that could indicate friction? ● Where do users stop or leave while viewing

		instructions? ● Do particular instruction areas appear to require more attention than others? (careful)
Direct, uncluttered access	Ensure recipe pages have a clear semantic structure, with descriptive headings and important recipe content easily accessible. Prioritize page performance and mobile usability, avoiding unnecessary content or elements that negatively affect loading speed and user experience.	● Can users easily find and navigate to the recipes they're looking for? ● Do users leave before reaching important recipe content? ● Are there repeated scrolling patterns, dead clicks or other interactions that may indicate navigation difficulties or unnecessary friction?
Images influence recipe selection	Use relevant, descriptive image alt text and ensure recipe images are included in Recipe JSON-LD and Open Graph metadata. Optimize	● How do users interact with recipe images and cards? ● Are certain recipe images/cards attracting

	images for loading performance without unnecessarily sacrificing quality.	more engagement than others? (careful) ● Do users click recipe images expecting them to be interactive?

Client Requirements

Area	Client Requirement	Planned Sprint	Outcome
Recipe structured data	Implement Schema.org/Recipe JSON-LD covering cooking times, ingredients, images and ratings	Sprint 2	Implemented and validated (Rich Results Test, 21 Oct.) Ratings not included: The Noroff API has no rating data and ratings were not invented.
Dynamic metadata	Generate recipe-specific page titles and meta descriptions	Sprint 2	Implemented and verified in production (22 Oct)
Open Graph	Generate recipe-specific social sharing metadata, including `og:title` and `og:image`	Sprint 2	Implemented and verified in production and with an external

			validator (22 Oct)
Lighthouse	Achieve 90+ in Performance, Accessibility and SEO	Sprint 3	Desktop met (98 / 97 /100). Mobile Accessibility 97 and SEO 100 met. Mobile Performance not met: average 86.3 (range 81-92) See detailed Lighthouse report (seo/lighthouse-audit.md)

Dynamic Metadata & Open Graph

Next.js provides the `generateMetadata()` function for generating page-specific metadata dynamically.

For FlavorCraft, this is useful because each recipe contains different information, such as title, description and image. Instead of using generic metadata across all recipe pages, `generateMetadata()` can use data fetched from the Noroff Recipes API.

Metadata to Generate

- Page title: Recipe title combined with the FlavorCraft brand
- Meta description: A description specific to the recipe
- Open Graph title: The recipe title for social sharing
- Open Graph description: A recipe-specific description for social sharing
- Open Graph image: The recipe image provided by the API

Planned implementation & Testing (Sprint 2)

- Use `generateMetadata()` in the dynamic recipe route.
- Retrieve metadata values from the Noroff Recipe API
- Provide appropriate fallback metadata when a recipe is missing a description or image.
- Verify that individual recipe pages generate the correct titles, descriptions and Open Graph tags.

Recipe Structured Data (JSON-LD)

What is Recipe JSON-LD?

Schema.org provides a standardized vocabulary for describing webpage content. JSON-LD is a format used to represent this information in a machine-readable way.

For FlavorCraft, we will use Schema.org/Recipe structured data to help search engines understand recipe information, such as ingredients, cooking times, instructions and images.

Correctly implemented structured data can make recipe pages eligible for enhanced search results, but it does not guarantee higher rankings or rich results.

Mapping the Noroff API to Schema.org

Noroff API Field	**Schema.org Property**	**Transformation**
`title`	`name`	Use directly
`description`	`description`	Use directly
`image.url`	`image`	Use the image URL
`prepTime`	`prepTime`	Convert minutes to ISO 8601 duration, eg. `PT20M`

`cookTime`	`cookTime`	Convert minutes to ISO 8601 duration, eg. `PT15M`
`prepTime + cookTime`	`totalTime`	Calculate total duration when appropriate
`servings`	`recipeYield`	Represents the number of servings
`ingredients`	`recipeIngredient`	Combine quantity, unit and name into readable strings
`instructions`	`recipeInstructions`	Convert each instruction into a `HowToStep` object

Planned Implementation & Testing (Sprint 2)

- Generate recipe-specific JSON-LD using the Noroff Recipes API
- Transform API fields into the formats required by Schema.org
- Ensure structured data matches the visible recipe content
- Handle missing or incomplete API data without inventing information
- Validate structured data using Google's Rich Results Test and Schema Markup Validator.

Implementation Progress — 21 October 2026

The planned Sprint 2 SEO features have progressed from research and planning to implementation and testing.

Requirement	Progress
Recipe structured data (JSON-LD)	Implemented, merged into `develop`, and validated on deployed recipe pages using Google's Rich Results Test.
Dynamic recipe metadata	Implemented, merged and verified on the deployed site, including missing-description fallback behavior.
Open Graph metadata	Implemented, merged and verified on the deployed site, including recipe-specific images and missing-image handling.
Lighthouse auditing	Initial baseline and follow-up performance investigations completed. Optimization findings communicated to the development team.

Detailed validation results, screenshots, and testing limitations are documented in **Tab 6 — Lighthouse & Technical SEO QA**.

References

- https://nextjs.org/docs/app/api-reference/functions/generate-metadata
- Schema.org Recipe: https://schema.org/Recipe
- Google Search Central – Recipe Structured Data: https://developers.google.com/search/docs/appearance/structured-data/recipe
- Google Rich Results Test: https://search.google.com/test/rich-results
- Schema Markup Validator: https://validator.schema.org/

Analytics Tool Selection

Selected Tool: Microsoft Clarity

Microsoft Clarity is the preferred analytics tool for FlavorCraft Labs.

Reasons for Selection

- Provides heatmaps to examine click and scroll behavior
- Offers session recordings to investigate navigation patterns and potential usability issues
- Includes insights into dead clicks and rage clicks
- Provides a free analytic solution suitable for the project's limited timeframe
- Supports custom events for tracking specific interactions where needed

Limitations & Considerations

- Behavioral data cannot reliably explain user intention without additional context.
- Some measurements, such as Favorites interactions, may require custom event tracking
- Privacy, consent requirements and data collection settings must be reviewed before implementation
- The team will need sufficient test activity to collect meaningful observations during Sprint 2.

Planned Implementation

Integrate Microsoft Clarity during Sprint 2, collect behavioral observations and use relevant findings to propose UX improvements during Sprint 3.

Reference:

https://clarity.microsoft.com/

Implementation Outcome — 21 October 2026

Microsoft Clarity was successfully integrated into FlavorCraft Labs during Sprints 2–3.

The implementation includes:

- Consent-aware analytics tracking.
- Four custom events: `recipe_favorited`, `recipe_unfavorited`, `recipe_searched`, and `recipe_opened_from_search`.
- Session recordings used to support usability testing.
- Manual verification of analytics consent acceptance, rejection, and withdrawal behavior.
- Review of the website's Privacy Policy disclosures.

All four custom events were confirmed as available in the Microsoft Clarity dashboard.

Six confirmed tester recordings were reviewed alongside seven questionnaire responses to identify potential usability improvements.

Status: Implementation and documented testing completed.

Detailed implementation and consent verification results are available in **Tab 4 — Analytics Measurement Plan & Implementation**.

Usability findings and recommendations are documented in **Tab 8 — Usability Testing & Analytics Findings**.

Custom Event Tracking

Proposed Custom Events

Event	**Trigger**	**Purpose**
recipe_favorited	A recipe is successfully saved to Favorites	Measure recipe-saving activity
recipe_unfavorited	A recipe is successfully removed from Favorites	Measure how often visitors remove saved recipes
recipe_searched	A search is executed	Understand how frequently visitors

		use search
recipe_opened_from_search	A visitor opens a recipe from search results	Understand whether visitors select recipes discovered through search

Privacy & Consent Considerations

Microsoft Clarity collects behavioral interaction data, including clicks, scrolling, page navigation and information used for session recordings.

Clarity uses cookies and identifiers to associate interactions with users and sessions. For visitors in the EEA, including Norway, valid consent signals are required for full cookie-based functionality.

Planned approach (Sprint 2)

- Obtain explicit analytics consent before activating Clarity tracking
- Provide clear options to accept, reject and withdraw consent
- Review masking settings to protect sensitive information
- Test that tracking respects users' consent choices
- Document analytics usage in the website's privacy information.

Considerations:
Without consent, come Clarity features may be limited. A consent-aware implementation may therefore reduce the amount of behavioral data available for Sprint 3.

References:

- https://learn.microsoft.com/en-us/clarity/setup-and-installation/consent-management
- https://learn.microsoft.com/en-us/clarity/setup-and-installation/clarity-consent-api-v2
- https://nkom.no/internett/informasjonskapsler-cookies

Original planning dependency: Clarification of public access and the appropriate analytics consent approach was identified before implementation. Consent-aware tracking has since been implemented and manually tested.

Analytics Measurement Plan

Planning note: The following measurement plan was prepared before the analytics tool and final tracking implementation were selected. The Microsoft Clarity implementation and verification report further down documents what was subsequently implemented and tested.

Objective

The purpose of analytics in FlavorCraft is to understand how users discover, navigate, and interact with recipes, and to identify potential usability issues.

Behavioral data, including clicks, scrolling patterns, and navigation paths, will be used to evaluate whether the application supports the needs identified through customer interviews.

The findings will help the team make evidence-informed UX improvements during sprint 3.

Scope

The initial measurement plan will focus on:

- Recipe discovery and search
- Category and dietary filtering
- Recipe selection and navigation

- Access to ingredients and instructions
- Favorites functionality
- Potential usability friction

Specific measurements and tracking methods will be finalized once interface and analytics tool have been selected.

Client Requirements

Behavioral analytics	Integrate a tool to observe clicks, scroll depth and navigation bottlenecks	Sprint 2
Data-driven improvements	Use session-recording insights to inform UI refinements	Sprint 3

Recipe Discovery - Measurement Plan

What we want to understand	Potential measurement	What it might tell us	
Are users finding recipes?	Percentage of feed visits that lead to opening a recipe, including visits involving search or filters	Whether users are progressing from discovery to viewing recipes	
Are search and filters helping users discover recipes?	Recipe-opening rates after searches of filter interactions	Whether users who use discovery tools go on to explore recipes	
Are users interested in the recipes they open?	Time spent on recipe pages, immediate returns to the feed, and scroll depth	Very short visits or repeated immediate returns might suggest the recipe didn’t meet	

		expectations	
Are users finding recipes worth saving?	Number/rate of recipes added to favorites after opening them	Whether users are engaging with recipes beyond simply viewing them	

Recipe Engagement - Measurement Plan

What we want to understand	**Potential measurement**	**What it might tell us**	
How do users navigate recipe content?	Scroll depth and repeated upward/downward scrolling	Whether users navigate smoothly or show patterns that might indicate difficulty finding information	
How do users move between ingredients and instructions	Repeated navigation between these sections observed through session recordings	Whether users frequently need to reference ingredients while following instructions, potentially suggesting opportunities	

		to improve the layout	
Are users reaching important recipe content?	Percentage of users reaching the ingredients and instructions sections	Whether essential information is accessible and being viewed	

Favorites - Measurement Plan

What we want to understand	**Potential measurement**	**What it might tell us**	
Do users return to their Favorites?	Number and frequency of visits to the Favorites page	Whether users return to their saved recipe collection	
Do users revisit saved recipes?	Recipe openings from the Favorites page	Whether users use Favorites to access previously saved recipes	
Do users manage their saved recipes?	Recipes added to and removed from Favorites	How users interact with their saved collection over time	

UX Friction - Measurement Plan

What we want to understand	Potential measurement	What it might tell us	
Are users struggling to find information?	Repeated scrolling up and down	Could indicate difficulty location information or a need to reference content in different sections	
Are users attempting to interact with non-interactive elements?	Dead clicks	Could reveal misleading visual elements or unclear interaction patterns	
Are users experiencing navigation difficulties?	Frequent back-and-forth navigation between pages	Could indicate confusing navigation or difficulty finding relevant recipes	
Are users experiencing unresponsive interactions?	Rage clicks	Could indicate broken functionality, slow responses or unclear feedback	

Implementation note:

This is an initial measurement plan based on customer research. Specific tracking methods, events and priorities will be finalized after the design and analytics tool have been selected. Findings will be interpreted alongside qualitative research, as behavioral patterns alone do not establish user intent.

Microsoft Clarity — Implementation & Testing Report

Implementation period: Sprints 2–3
Analytics tool: Microsoft Clarity
Testing date: 21 October 2026
Status: Implemented, merged, and tested

1. Implementation Overview

Microsoft Clarity was integrated into FlavorCraft Labs to collect behavioral analytics and support usability evaluation.

The implementation includes consent-aware analytics initialization and custom event tracking for recipe-related interactions.

Clarity provides session recordings, click and scroll observations, and additional insights into how visitors interact with the application.

2. Consent Management

Clarity was integrated with the application's analytics consent mechanism.

The implementation is designed to:

- Activate analytics tracking only after analytics consent has been granted.
- Allow visitors to reject analytics tracking.
- Allow visitors to change or withdraw their consent.
- Prevent custom analytics events from being sent without analytics consent.

Consent behavior was manually tested using Chrome DevTools.

Consent Verification — 21 October 2026

Testing was performed using the Network tab in Chrome DevTools, filtered for Microsoft Clarity requests.

Scenario	Observation	Result
Reject analytics consent	No Clarity network requests detected after rejection and navigation	Passed

Accept analytics consent	Multiple Clarity `collect` requests observed with HTTP 204 responses	Passed
Withdraw analytics consent	No new Clarity tracking requests observed after withdrawal and further interaction	Passed

These observations support that analytics tracking respects visitor consent choices in the tested scenarios.

The withdrawal test was based on observing new network activity after consent was withdrawn. It does not independently establish whether previously stored analytics identifiers or cookies were removed.

Privacy Information Verification — 21 October 2026

The deployed application includes a Privacy Policy explaining the use of Microsoft Clarity, the purpose of behavioral analytics, the types of information collected, and the use of cookies and local storage.

The policy also explains that analytics consent is optional and that visitors can accept, reject, or withdraw consent through Privacy Settings.

The Privacy Policy was manually reviewed and confirmed to include these disclosures.

Status: Privacy information present and reviewed. Technical verification of cookie deletion after withdrawal was not included in this review.

3. Custom Event Tracking

Four custom events were implemented to measure recipe-related interactions.

Event	Trigger	Purpose
`recipe_searched`	A visitor executes a recipe search	Understand search activity
`recipe_opened_from_search`	A visitor opens a recipe from search results	Understand recipe selection following search
`recipe_favorited`	A recipe is successfully added to Favorites	Measure recipe-saving activity
`recipe_unfavorited`	A recipe is successfully removed from Favorites	Measure removal of saved recipes

The events were tested during development and confirmed as available in the Microsoft Clarity dashboard after deployment.

Recorded sessions were also observed for `recipe_searched`, `recipe_opened_from_search`, and `recipe_favorited`.

4. Testing & Verification

Test	Result
Clarity initializes after analytics consent	Passed
No Clarity requests detected after rejecting consent	Passed
Clarity requests detected after accepting consent	Passed

No new tracking requests observed after withdrawing consent	Passed
Recipe search event	Passed
Recipe opened from search event	Passed
Recipe favorited event	Passed
Recipe unfavorited event	Passed
All four custom events available in Clarity dashboard	Passed
Session recordings collected	Passed

5. Analytics Data Collection

Microsoft Clarity was used to review visitor interactions during usability testing.

Six confirmed tester recordings were manually reviewed alongside seven questionnaire responses.

Additional recordings that could not be reliably attributed to testers were excluded from the verified usability analysis.

The recordings were used to investigate navigation behavior, recipe discovery, and potential usability friction.

Detailed findings and recommendations are documented in **Tab 10 — Usability Testing & Analytics**.

6. Limitations

- Analytics collection depends on visitor consent, which reduces the available sample.
- Session recordings cannot reliably establish user intention or satisfaction.
- Automated dead-click and rage-click warnings require manual verification.
- Development activity may appear in analytics unless appropriately excluded.
- The limited testing period and small sample size restrict the conclusions that can be drawn.
- Custom event availability does not necessarily establish that every event occurred frequently during usability testing.

7. Evidence

The following screenshots document implementation and consent verification.

Figure 1: Custom event availability

Microsoft Clarity event filter showing all four implemented custom events, including `recipe_unfavorited`.

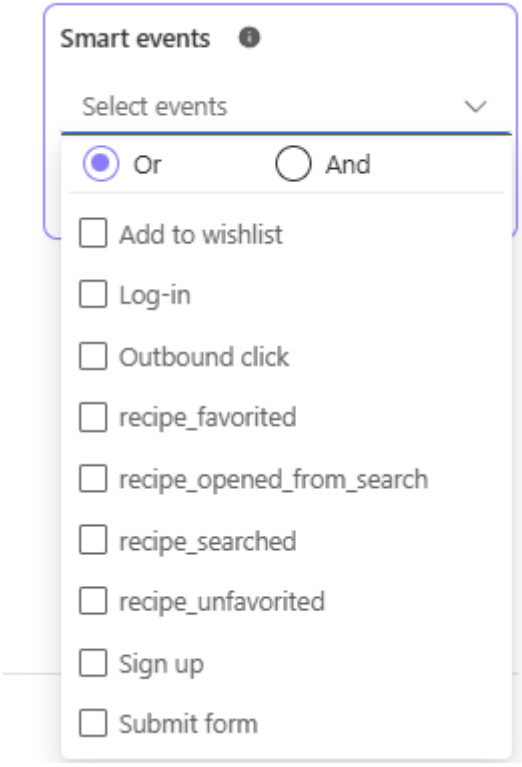


Figure 2: Recorded custom events

Microsoft Clarity dashboard showing recorded sessions associated with `recipe_searched`, `recipe_opened_from_search`, and `recipe_favorited`.

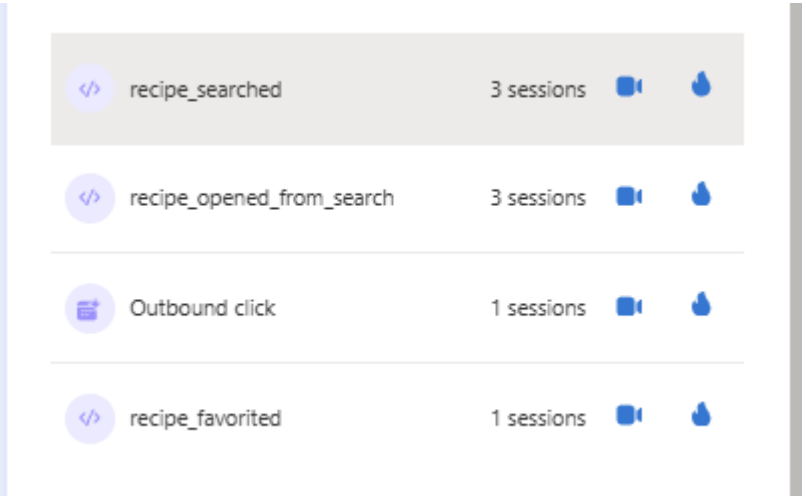


Figure 3: Consent rejection

Chrome DevTools Network tab showing no Microsoft Clarity requests detected after rejecting analytics consent and navigating the application.

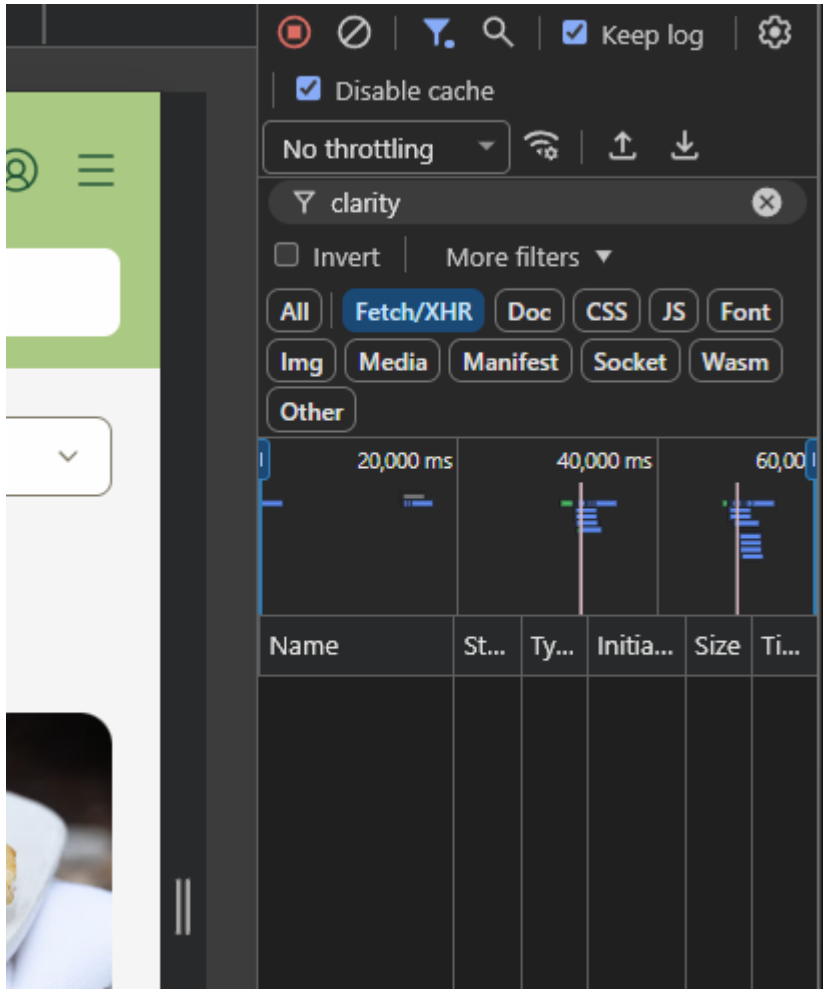


Figure 4: Consent acceptance

Chrome DevTools Network tab showing Microsoft Clarity `collect` requests after analytics consent was granted, with HTTP 204 responses.

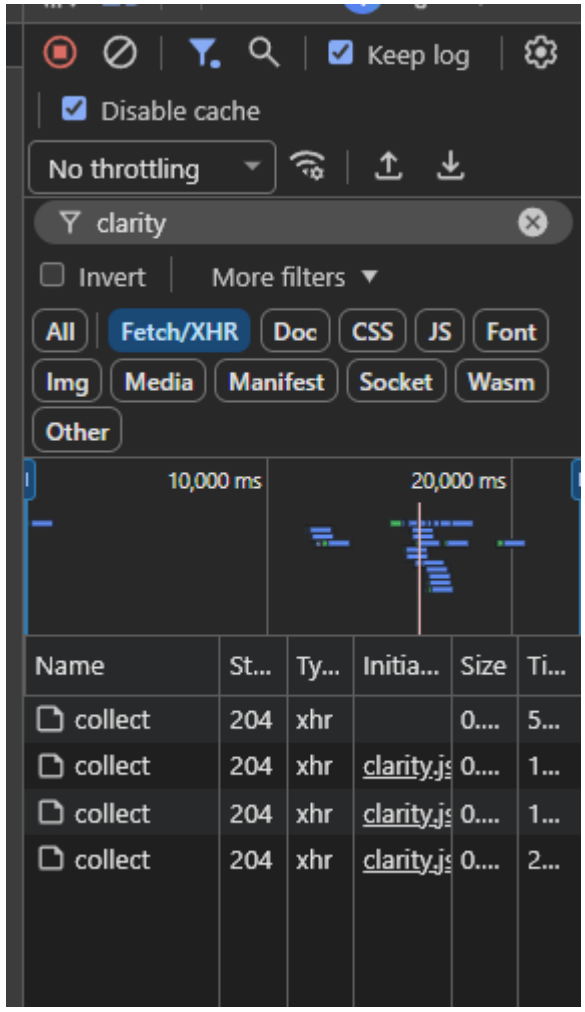
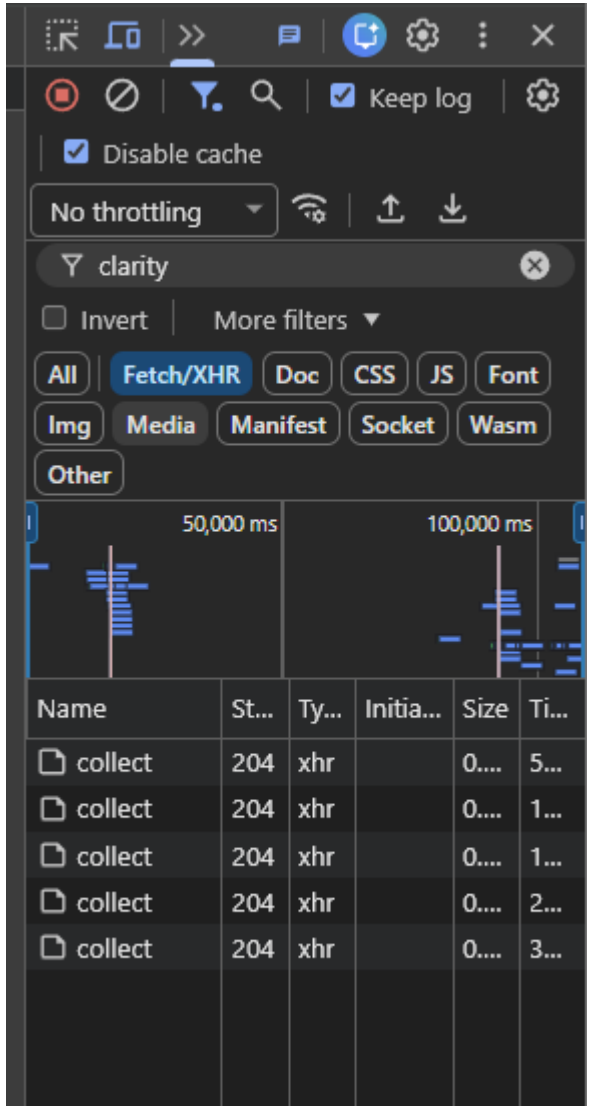


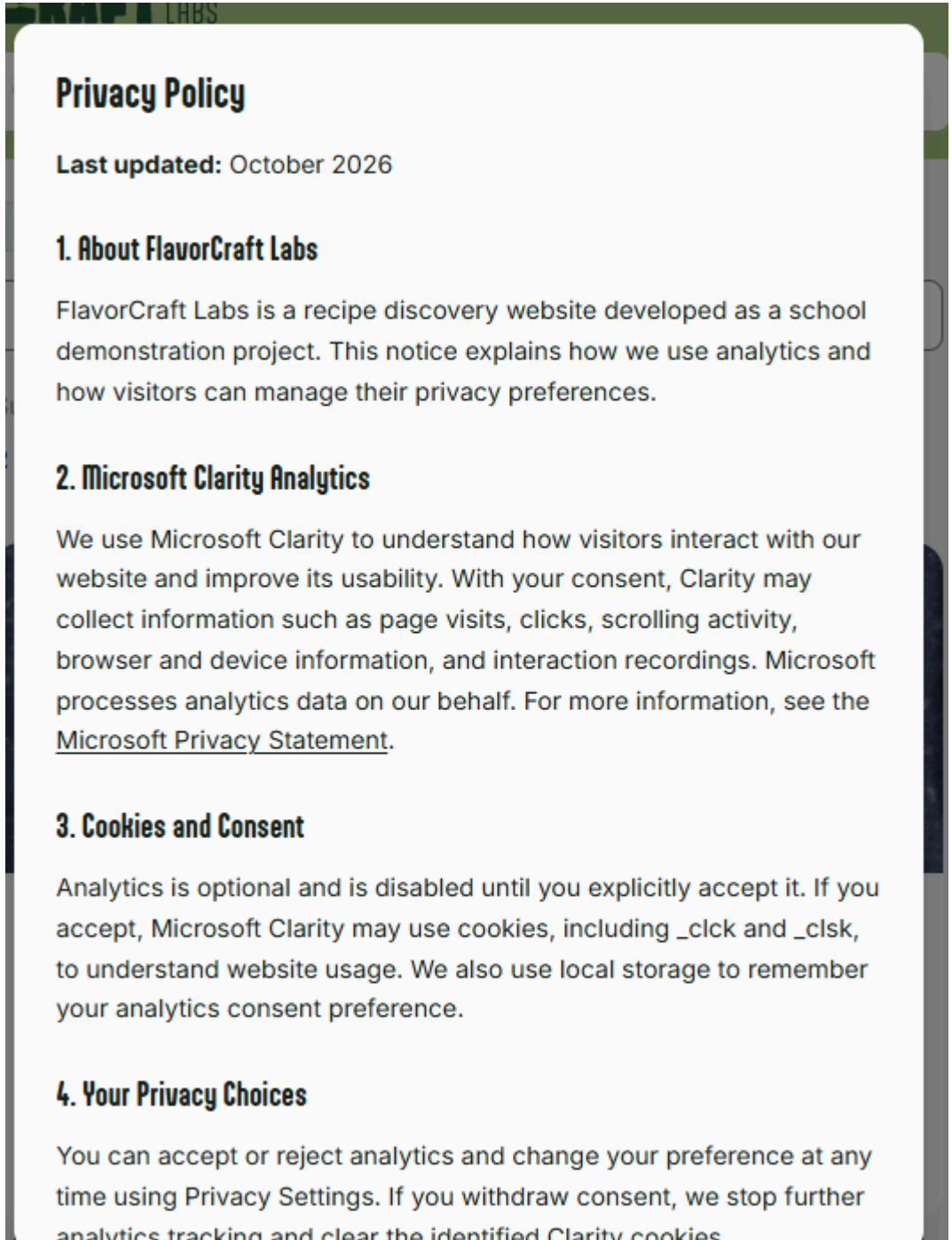
Figure 5: Consent withdrawal

Chrome DevTools Network tab used to verify that no new Microsoft Clarity tracking requests were observed after withdrawing analytics consent and continuing to interact with the application.



Additional evidence is available through the Microsoft Clarity session recordings and the usability testing report.

Figure 6: Privacy Policy



Privacy Policy

Last updated: October 2026

1. About FlavorCraft Labs

FlavorCraft Labs is a recipe discovery website developed as a school demonstration project. This notice explains how we use analytics and how visitors can manage their privacy preferences.

2. Microsoft Clarity Analytics

We use Microsoft Clarity to understand how visitors interact with our website and improve its usability. With your consent, Clarity may collect information such as page visits, clicks, scrolling activity, browser and device information, and interaction recordings. Microsoft processes analytics data on our behalf. For more information, see the Microsoft Privacy Statement.

3. Cookies and Consent

Analytics is optional and is disabled until you explicitly accept it. If you accept, Microsoft Clarity may use cookies, including _clck and _clsk, to understand website usage. We also use local storage to remember your analytics consent preference.

4. Your Privacy Choices

You can accept or reject analytics and change your preference at any time using Privacy Settings. If you withdraw consent, we stop further analytics tracking and clear the identified Clarity cookies.

8. Overall Status

Microsoft Clarity integration, custom event tracking, and the tested consent scenarios have been implemented and verified.

Behavioral data was successfully collected and used to support the usability evaluation.

Implementation status: Complete
Testing status: Complete for the scenarios documented above
Documentation status: Privacy information reviewed and documented. Technical verification of cookie deletion after consent withdrawal remains outside the scope of the completed tests.

Sprint 1 - Lighthouse Baseline Status

Initial Status: Blocked - awaiting a testable deployment/

The team has configured Netlify deployment URLs for the `main` and `develop` branches. However, no complete pages are currently available for testing.

A meaningful baseline Lighthouse audit cannot be performed at this stage.

Next step: Run and document the initial Lighthouse audit once a working page is deployed and accessible.

Dependency: Development team / Testable deployment

Current status, 20 October: Unblocked. A testable deployment became available, and Lighthouse audits were subsequently performed. Findings and recommendations are documented in the Lighthouse audit.

Detailed Lighthouse baseline results, performance investigations, accessibility findings, and recommendations are maintained in `src/app/docs/seo/lighthouse-audit.md`.

Recipe Structured Data (JSON-LD) — Validation

Recipe JSON-LD was implemented dynamically using data from the Noroff Recipes API and merged into `develop`.

The JSON-LD script is rendered in the page body, rather than `<head>`. Google supports JSON-LD anywhere in the page, and the Rich Results Test detected it on both pages.

On 21 October 2026, two deployed recipe pages were tested using Google's Rich Results Test:

- **Banana Split 2:** 1 valid Recipe item detected; 10 non-critical warnings.
- **Banana Smoothie:** 1 valid Recipe item detected; 16 non-critical warnings.

Both pages passed validation and were identified as eligible for recipe-rich results.

The reported warnings concern optional structured data properties. Missing information was not fabricated to eliminate these warnings.

Status: Implementation, deployment verification, and validation completed.

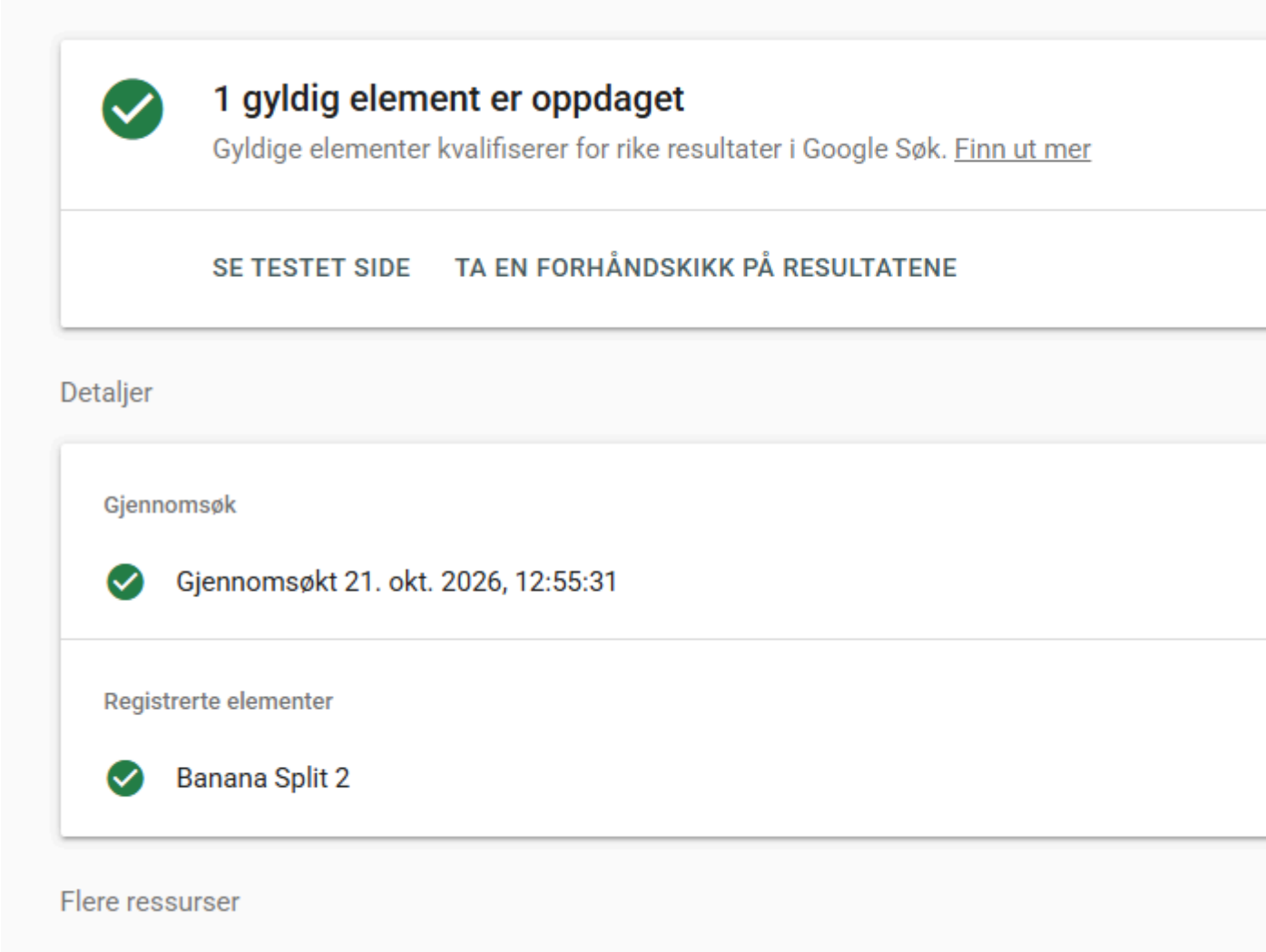


Figure 1: Google Rich Results Test - Banana Split 2, one valid Recipe item detected, 21 October

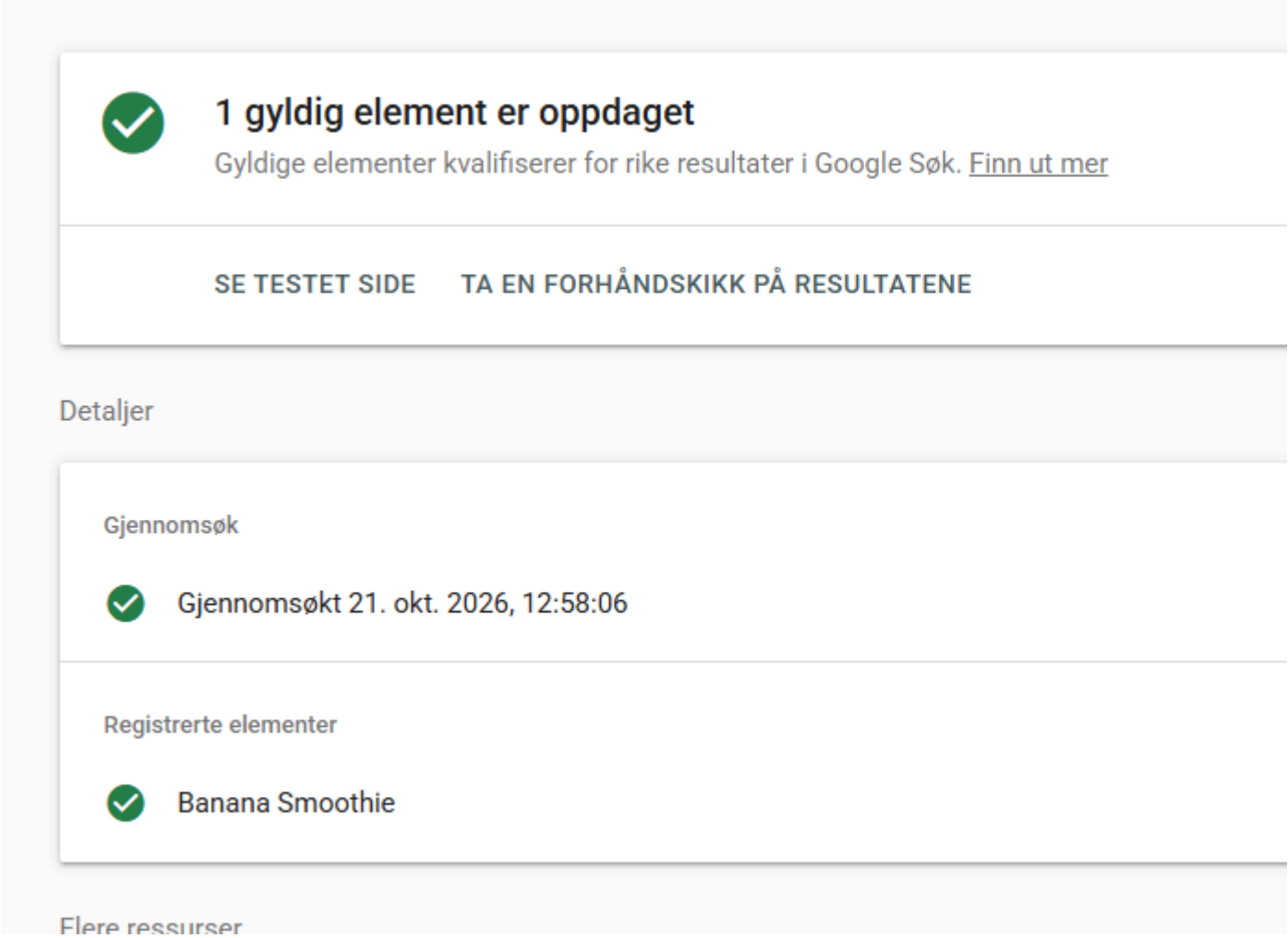


Figure 2: Google Rich Results Test - Banana Smoothie, one valid Recipe item detected, 21 October

Dynamic Recipe Metadata & Open Graph — Validation

Ticket: #50 — Dynamic Recipe Metadata & Open Graph
Testing dates: 21–22 October 2026
Environment: Local development and deployed Netlify site
Status: Implementation merged and deployed. Local testing and

production metadata verification passed. External preview validation completed.

Testing performed

The implementation uses Next.js `generateMetadata()` to dynamically generate recipe-specific SEO and Open Graph metadata using data from the Noroff Recipes API.

Test case	Result
Recipe-specific page title	Passed
Recipe-specific meta description	Passed
Recipe-specific Open Graph title and description	Passed
Recipe-specific Open Graph image	Passed
Missing description	Passed — fallback description generated
Missing image	Passed — `og:image` omitted without breaking the page
Metadata present in rendered HTML	Passed

Test observations

Beef Wellington: Confirmed that the page title, meta description, Open Graph title, description, image, and image alternative text contained recipe-specific information. The image URL was also manually verified as accessible.

Missing description: Confirmed that both the standard meta description and Open Graph description used the fallback: "Discover this delicious recipe on FlavorCraft Labs."

Missing image: Confirmed that a recipe without an image did not generate an `og:image` tag and that the page continued to render correctly.

A separate recipe contained an image URL pointing to an Imgur page rather than a direct image resource. This is a potential external-data quality limitation, not a failure of the missing-image fallback.

Production verification — 22 October 2026

Following the merge and deployment, verified that dynamic Open Graph metadata is present on the deployed Netlify site.

Inspected individual recipe pages and confirmed that the generated metadata contains the correct recipe-specific information.

This confirms that the dynamic metadata implementation is working in production, rather than only in the local development environment.

Evidence

Screenshots captured from Chrome DevTools showing recipe-specific metadata and fallback behavior.

External Open Graph validation — 22 October 2026

The deployed Greek Salad recipe was tested using an external Open Graph preview validator.

Passed:

- Recipe-specific Open Graph title and description detected.
- Open Graph image successfully retrieved (212 KB, JPEG).
- Page title correctly includes the recipe name and FlavorCraft Labs.
- X/Twitter large-image card metadata detected, including title, description, and image.

Improvement opportunities identified:

- The source image dimensions (800 × 1200) differ from the recommended landscape aspect ratio for social sharing (1200 × 630), which may cause cropping or letterboxing.

- The recipe description is shorter than the validator's recommended length for search snippets.
- The validator recommended promotional text within the image; this is optional and not required for valid Open Graph metadata.

Conclusion: The external validator successfully retrieved the recipe-specific Open Graph metadata and image from the deployed application. The remaining findings are optimization opportunities rather than evidence that metadata generation has failed.

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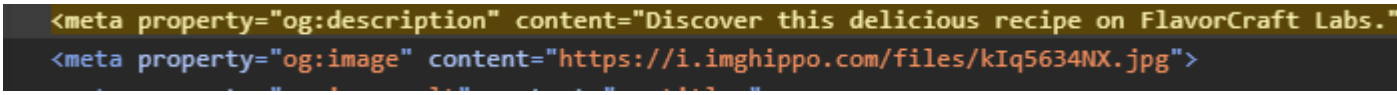


Figure 3: Missing recipe description — fallback metadata generated successfully (localhost).

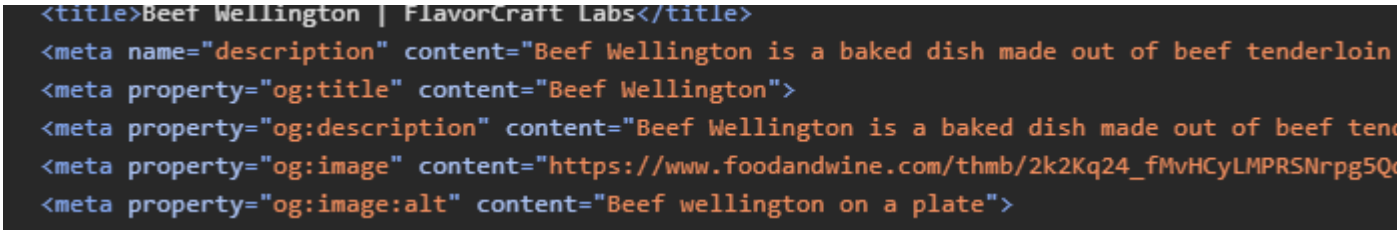


Figure 4: Beef Wellington — recipe-specific title, description, and Open Graph image metadata (localhost).

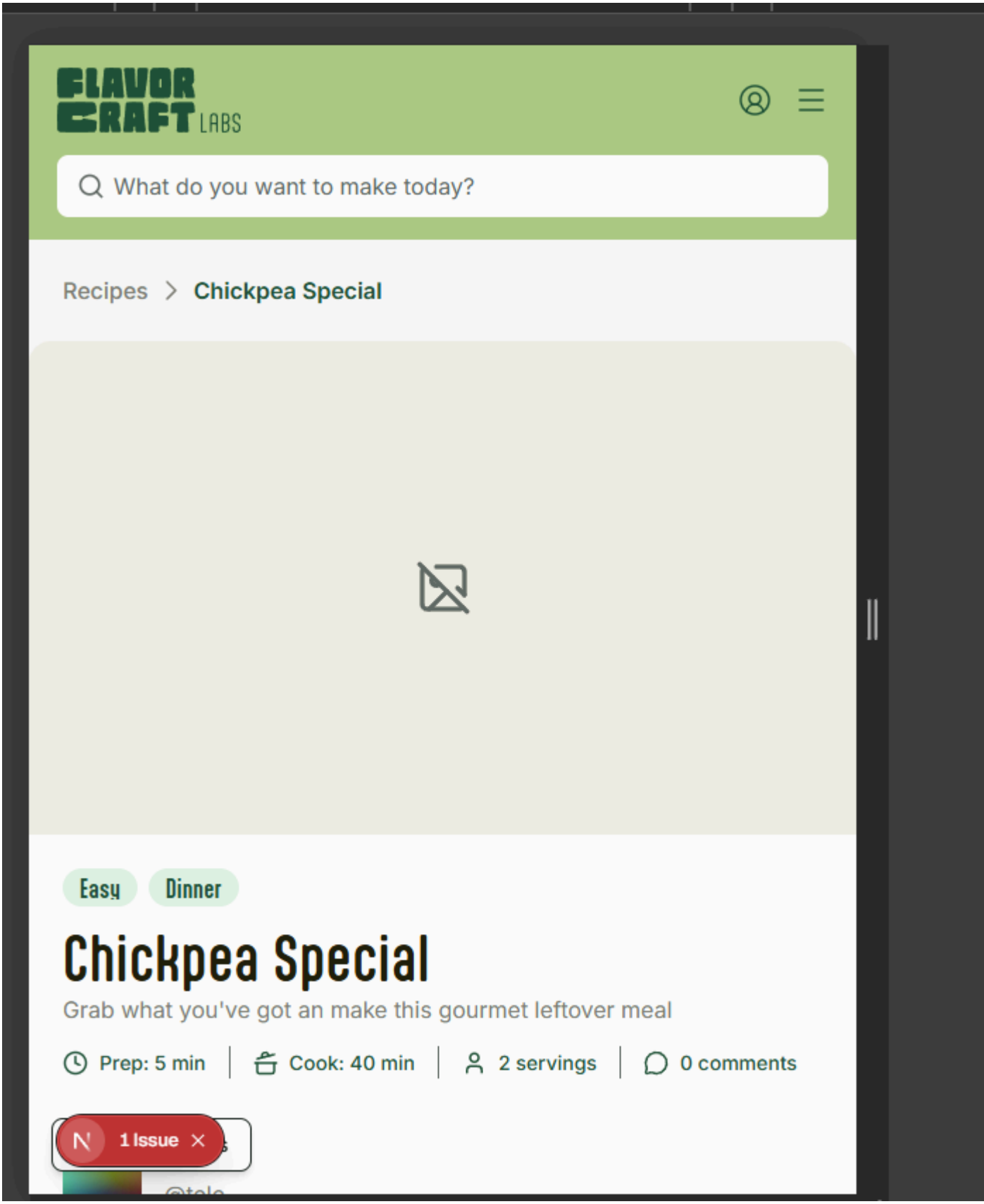


Figure 5: Missing recipe image — `og:image` correctly omitted (localhost).

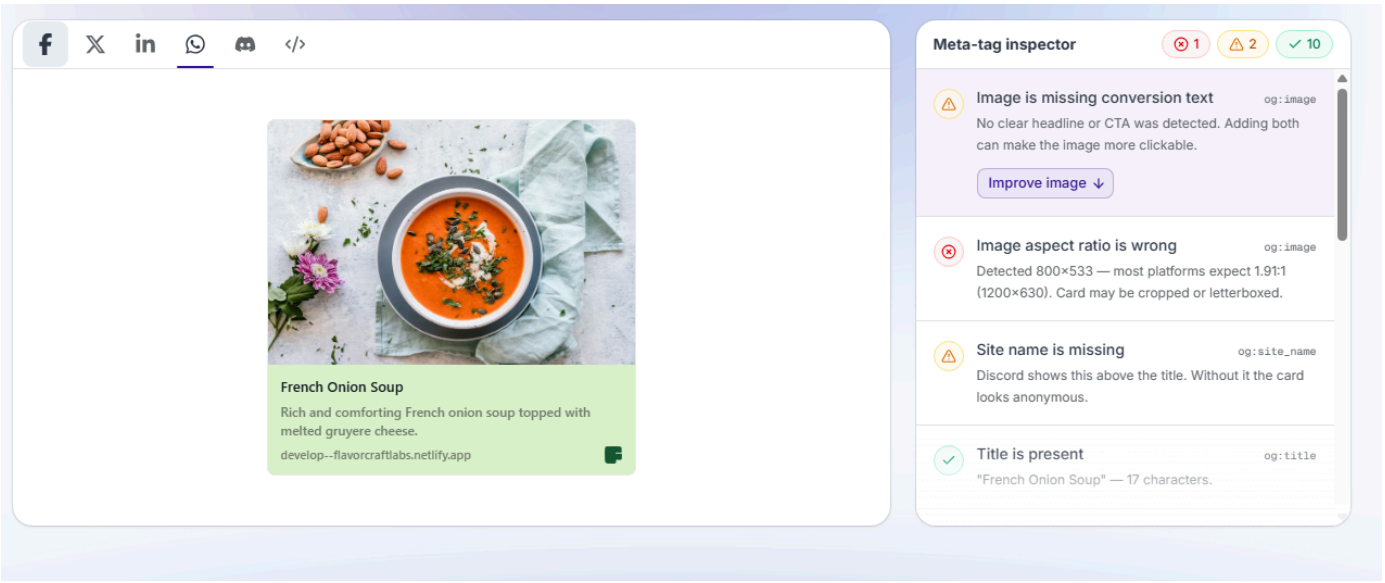


Figure 6: https://www.opengraph.xyz/ test of recipe detail url

Conclusion: Dynamic recipe metadata has been successfully implemented, merged, deployed, and verified on individual recipe pages.

These instructions were prepared for usability testing conducted during Sprint 3. Two versions were provided: one for participants assigned a test account and one for visitors testing without authentication. Both versions included comparable recipe discovery and navigation tasks, while authenticated testers were also asked to evaluate Favorites functionality.

Test Guide A — Authenticated User

Thank you for helping us test FlavorCraft Labs!

FlavorCraft Labs is a recipe application developed by Front-end Development students at Noroff. We're looking for feedback on how easy the application is to use and whether anything could be improved.

Estimated testing time: 10–15 minutes

Website: https://develop--flavorcraftlabs.netlify.app/

1. Your Test Account

We've created a test account for you. You do not need to register.

- **Email:** flavorcraft.test03@stud.noroff.no
- **Password:** **********

Please use only the account assigned to you. Do not enter personal passwords or sensitive information.

2. Testing tasks

Please complete the following tasks without assistance if possible. We're interested in how intuitive the application is, so there are no right or wrong answers.

Task 1 — Explore the homepage

Visit the homepage and explore the available recipes. Is it clear what the website offers and where to begin?

Task 2 — Find a recipe

Imagine you're looking for something to cook for dinner. Use the search function or category filters to find a suitable recipe.

Task 3 — View recipe details

Open a recipe that interests you. Find its ingredients and cooking instructions. Consider whether the information is clear and easy to follow.

Task 4 — Log in and save a recipe

Log in using your assigned test account. Find a recipe you like and save it to your favorites, if that feature is available.

Task 5 — Explore freely

Spend a few minutes navigating the website. Try anything that interests you and note any difficulties or unexpected behavior.

You may accept or reject optional analytics tracking. Your decision will not affect your participation.

3. Feedback

Once you've completed the tasks, please answer the following questions.

1. How easy was the website to navigate? (1 = Very difficult, 5 = Very easy)
2. How easy was it to find a recipe? (1–5)
3. Was the recipe information clear and easy to understand?
4. Did you encounter any errors, broken features, or confusing elements?
5. What would you improve about the website?

If you encountered an issue, please describe what you were trying to do when it happened.

Device and browser: [e.g., iPhone/Safari or Windows/Chrome]

Send completed form to: Whoever recruited you or *****@stud.noroff.no

Thank you for taking the time to help us improve FlavorCraft Labs!

Test Guide B — Visitor (Not Logged In)

Thank you for helping us test FlavorCraft Labs!

FlavorCraft Labs is a recipe application developed by Front-end Development students at Noroff. We'd love your feedback before our final submission.

Testing website: https://develop--flavorcraftlabs.netlify.app/

Estimated time: 10–15 minutes

1. Getting started

You do not need to register or log in to complete this test.

Simply open the website using the link above and explore it as a visitor.

2. Testing tasks

Please complete the following tasks without assistance if possible. We're interested in how intuitive the application is, so there are no right or wrong answers.

Task 1 — Explore the homepage

Visit the homepage and explore the available recipes. Is it clear what the website offers and where to begin?

Task 2 — Find a recipe

Imagine you're looking for something to cook for dinner. Use the search function or category filters to find a suitable recipe.

Task 3 — View recipe details

Open a recipe that interests you. Find its ingredients and cooking instructions. Consider whether the information is clear and easy to follow.

Task 4 — Explore freely

Spend a few minutes navigating the website. Try anything that interests you and note any difficulties or unexpected behavior.

You may accept or reject optional analytics tracking. Your decision will not affect your participation.

3. Feedback

Once you've completed the tasks, please answer the following questions.

1. How easy was the website to navigate? (1 = Very difficult, 5 = Very easy)
2. How easy was it to find a recipe? (1–5)
3. Was the recipe information clear and easy to understand?
4. Did you encounter any errors, broken features, or confusing elements?
5. What would you improve about the website?

If you encountered an issue, please describe what you were trying to do when it happened.

Device and browser: [e.g., iPhone/Safari or Windows/Chrome]

Send completed form to: Whoever recruited you or *****@stud.noroff.no

Thank you for taking the time to help us improve FlavorCraft Labs!

FlavorCraft Labs — Usability Testing Report

Project: FlavorCraft Labs
Date: 21 October 2026
Prepared by: SEO & Analytics Lead
Testing methods: User questionnaires and Microsoft Clarity session recordings

1. Executive Summary

Usability testing of FlavorCraft Labs showed generally positive results. Participants were able to discover recipes, browse categories, use search and filtering, and interact with recipe details.

The strongest areas were recipe discovery and exploration. However, feedback identified opportunities to improve the clarity of filtering, recipe categorization, and interaction feedback.

The testing also highlighted the importance of distinguishing genuine usability issues from automated analytics warnings. Several interactions identified as dead clicks by Microsoft Clarity were manually reviewed and found to function correctly.

Overall assessment: The core recipe browsing experience is functional and discoverable. The main opportunities for improvement relate to usability refinements, content quality, and performance rather than fundamental navigation failures.

2. Testing Methodology

The usability evaluation combined two sources of evidence:

- **User questionnaires:** Seven responses collected from usability testers.

- **Microsoft Clarity:** Six confirmed tester recordings reviewed to observe navigation, filtering, recipe exploration, and other interactions.

Two additional Clarity recordings could not be reliably attributed to testers and were excluded from the verified session analysis.

Questionnaire feedback was treated as the primary source for participants' experiences and opinions. Clarity recordings were used as supporting evidence of observable behavior.

Limitations

- The sample size was relatively small.
- Not every questionnaire response included a numerical rating.
- Session recordings do not reliably reveal users' intentions or satisfaction.
- Clarity's automated dead-click detection produced several misleading flags.
- Some browser-native interactions, such as sharing dialogs, may not be visible in recordings.
- Clarity heatmap and scroll-depth data may include developer activity because internal traffic could not be reliably excluded.

The testing tasks and questionnaires provided to participants are documented in Tab 7 — Usability Testing — Participant Instructions.

3. Questionnaire Results

The feedback received was generally positive.

Area	Reported result
Recipe discovery	Average 5/5 from five numerical responses
Navigation	Average 4.5/5 from two numerical responses

Total questionnaire responses	7

These averages are descriptive only and should not be interpreted as statistically representative results.

Positive feedback

Participants generally found recipes easy to discover and were able to explore the available content.

Areas for improvement

Questionnaire feedback highlighted several concerns:

- **Filtering behavior:** Some participants found it unclear how multiple selected filters interacted.
- **Filter responsiveness:** Delays or unclear feedback when selecting filters were reported.
- **Results visibility:** Participants sometimes needed to scroll to locate updated results.
- **Recipe categorization:** Some recipes appeared in unexpected categories, such as brownies under Dinner or Beef Wellington appearing after dessert results.
- **Tags:** Some labels were difficult to interpret.
- **Recipe content:** Placeholder content, unsuitable imagery, and unfamiliar measurements reduced content clarity.
- **Navigation convenience:** A sticky search bar was suggested.
- **Incomplete features:** Meal Planner and Pantry were still temporary or unfinished sections during testing.

4. Microsoft Clarity Session Findings

Sessions 1–2: Recipe discovery and navigation

Both sessions showed visitors exploring recipes, using filters, opening recipe details, and navigating between sections.

Clarity flagged several dead clicks, but manual review showed that the relevant interactions, including Pantry navigation and removing a filter tag, worked as expected.

Assessment: No confirmed usability defects identified.

Session 3: Profile editing feedback

The visitor explored authentication, profile editing, recipe search, filtering, favorites, and sharing.

A click on **Edit Profile** appeared to produce no visible response on desktop. Further investigation established that the form was already displayed beside the button. On mobile, the same interaction scrolls the user to the form, providing clearer feedback.

Finding: The Edit Profile interaction lacks sufficiently obvious feedback on desktop.

Recommendation: Move keyboard focus to the first editable field and display a clear focus indicator. This improvement has already been suggested to the development team.

Session 4: Homepage navigation performance

The visitor successfully searched for pizza, combined filters, opened recipes, and interacted with ingredients and instructions.

Clicking the FlavorCraft logo returned the visitor to the homepage, but navigation appeared slow.

Finding: Noticeable homepage navigation delay.

Recommendation: Address through the existing Lighthouse performance investigation rather than creating a duplicate usability issue.

Session 5: Recipe browsing

The visitor navigated between recipe pages, used the Dinner filter, and opened additional recipes.

A dead-click warning on the logo was manually reviewed. Navigation worked correctly.

Assessment: No new confirmed usability issues.

Session 6: Recipe sharing and details

The visitor explored Beef Wellington and repeatedly interacted with the Share feature.

Clarity flagged clicks near the ingredients and instructions section, but visual review suggested the clicks were concentrated around the Share button.

A separate manual test confirmed that Share opens a sharing popup.

Assessment: No confirmed sharing defect. The recording alone cannot establish whether the browser's sharing interface appeared to the visitor.

5. Heatmaps and Scroll Depth

Microsoft Clarity heatmaps were reviewed to identify interaction patterns across the homepage, recipe listing page, and individual recipe pages. Scroll-depth data was also examined to understand how far visitors navigated through the available content.

Heatmap observations

Homepage

The homepage click heatmap shows interactions with the main recipe discovery button, navigation elements, category links, and recipe cards.

Clicks were also recorded further down the homepage, suggesting that some visitors explored recipe recommendations beyond the initial viewport.

Recipe listing page

The recipe listing heatmap shows interactions with navigation, search, filtering controls, and recipe cards.

These observations are consistent with the session recordings, where testers explored recipes through several different navigation paths.

Recipe detail page

The Beef Wellington recipe detail heatmap shows interactions around the ingredients and instructions controls, quantity adjustment buttons, and sharing functionality.

This suggests engagement with recipe-specific features beyond simply viewing the recipe information.

However, the heatmap alone cannot confirm whether every interaction was successful or whether users understood the functionality.

Scroll-depth observations

Microsoft Clarity's Overview dashboard reported an **average scroll depth of 71.37%** across the recorded activity included in the selected reporting period.

This suggests that visitors generally scrolled through a substantial portion of the pages they viewed.

However, the metric represents aggregated activity and does not indicate whether visitors read or engaged with the content.

Furthermore, because developer activity could not be reliably excluded, the reported average may include internal testing and development sessions.

Interpretation: The scroll-depth result provides a general indication of browsing behavior but should not be treated as a reliable measure of engagement among the verified usability testers alone.

Data reliability and limitations

The heatmaps and scroll-depth results must be interpreted cautiously because developer activity may have influenced the collected analytics data.

Before usability testing, the development team was asked to avoid using the deployed application or provide their IP addresses so that development activity could be excluded from the analytics.

However, IP addresses were not provided, and it was not possible to verify that all developers avoided the deployed application.

Consequently, some recorded interactions may originate from development or internal testing rather than external usability participants.

This introduces several limitations:

- Heatmap click concentrations may reflect repeated developer interactions rather than genuine visitor behavior.
- Average scroll depth may be affected by internal testing and debugging sessions.
- The available analytics data cannot be treated as a representative sample of independent users.
- Patterns observed in heatmaps cannot always be reliably attributed to the seven questionnaire participants or six verified tester recordings.

For these reasons, heatmaps and scroll-depth statistics were used as **supplementary, exploratory evidence**, rather than definitive measures of user behavior.

Greater weight was given to questionnaire feedback and manually reviewed, identifiable usability testing sessions.

Overall assessment

The heatmaps indicate interaction with key recipe discovery and detail-page features, while the scroll-depth data suggests that visitors explored content beyond the initial viewport.

These observations are broadly consistent with the questionnaire responses and verified session recordings.

However, possible contamination from developer activity limits the reliability of the aggregate analytics data. The findings should therefore

be interpreted as indicative patterns rather than conclusive evidence of usability or engagement.

Heatmaps

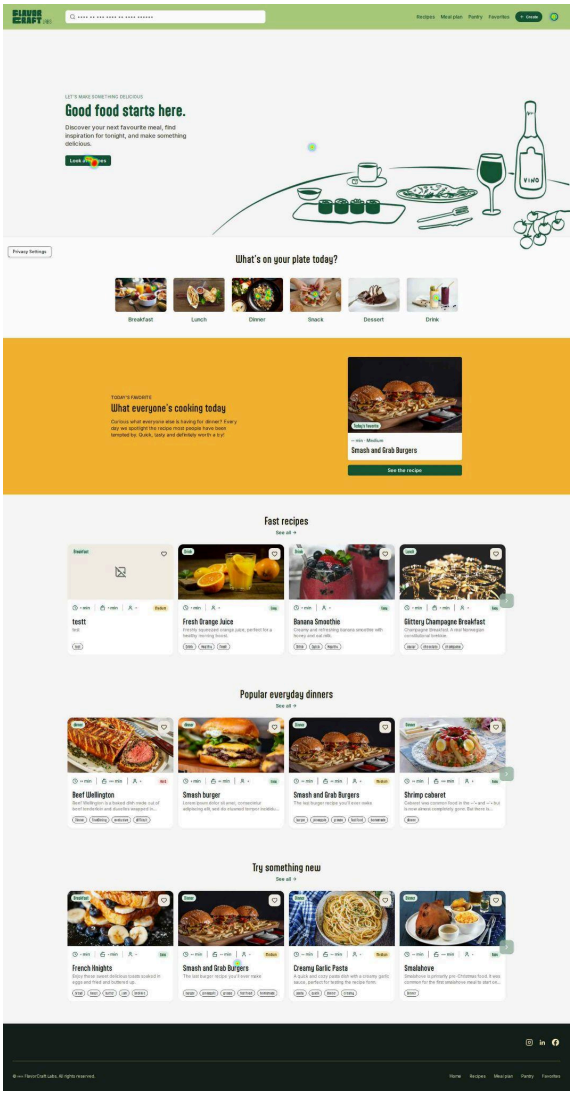


figure 1: Microsoft Clarity homepage click heatmap, showing interactions with navigation, recipe discovery, categories, and recipe recommendations (20 October 2026, 18.00 - 22.00).

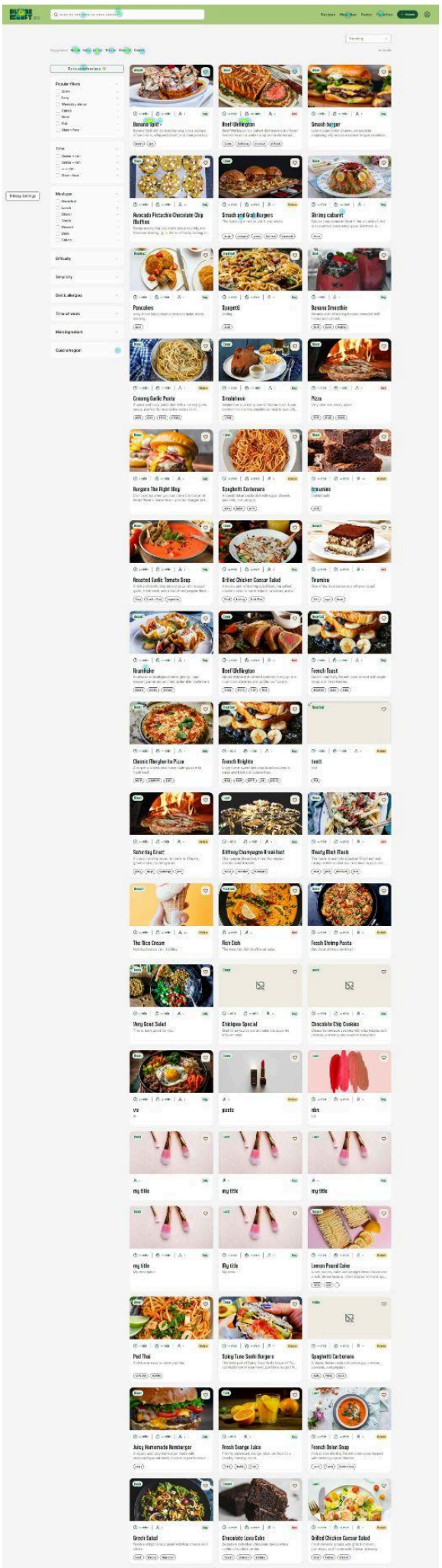


Figure 2: Microsoft Clarity recipe listing click heatmap, showing interactions with search, filtering controls, and recipe cards (20 October 2026, 18.00 - 22.00).

FLAVOR CRAFT LABS

Recipes > Beef Wellington



Hard dinner fineDining exclusive difficult

Beef Wellington

Beef Wellington is a baked dish made out of beef tenderloin and duxelles wrapped in shortcrust pastry.

Privacy Settings Cook: ▫▫▫ min ▫ servings ▫ comments

InspireInk
@InspireInk

Share Meal planner Print out Favorite

Ingredients Instructions

Ingredients

▫ pcs Ingredient placeholder.

Share

Find similar recipes

Hard dinner

Comments

You have to be logged in to comment

No comments yet. Be the first to share your thoughts.

Home
Recipes
Meal plan
Pantry
Favorites

Figure 3: Microsoft Clarity Beef Wellington recipe detail click heatmap, showing interactions with recipe controls and sharing functionality (20 October 2026, 18.00 - 22.00).



Figure 4: Microsoft Clarity Overview dashboard showing an average scroll depth of 71.37% for the selected reporting period (20 October 2026, 18.00 - 22.00).

6. Prioritized Findings and Recommendations

Priority	Finding	Recommendation
High	Unclear combined filter behavior	Clarify how selected filters interact and mak easier to understand
High	Unexpected recipe categorization	Review category and tag assignments and filtering accuracy
Medium	Delayed or unclear filter feedback	Improve responsiveness and communicate update
Medium	Updated results may be difficult to locate	Improve result visibility after filter changes
Medium	Slow homepage navigation	Investigate alongside Lighthouse performan
Low–Medium	Edit Profile lacks clear desktop feedback	Focus the first input and provide a visible fo
Medium	Placeholder or unsuitable recipe content	Review imagery, recipe text, and measurem presentation
Future improvement	Search bar visibility and unfinished features	Evaluate sticky search and complete Meal I in later iterations

Priorities reflect the available feedback and potential impact, not a formal severity assessment.

7. Analytics Observations

Microsoft Clarity was useful for observing navigation patterns and investigating specific interactions. However, automated warnings required manual validation.

Several flagged dead clicks were successful interactions, including logo navigation, filter removal, and Pantry navigation.

This demonstrates that analytics events should be interpreted in context rather than treated as confirmed defects.

The recordings also showed users engaging with recipe search, categories, favorites, ingredients, instructions, and sharing functionality.

8. Recommendations Before Submission

Given the remaining project time, the recommended focus is:

1. **Review filtering and recipe categorization**, prioritizing any clear functional or content errors.
2. **Improve Edit Profile feedback** if the change can be implemented and tested safely.
3. **Address critical content issues**, particularly misleading categories, placeholder text, or unsuitable images.
4. **Document remaining performance limitations** and link them to the Lighthouse audit.
5. **Retest any implemented changes** to ensure they do not introduce regressions.

Larger architectural performance improvements and unfinished feature development should be evaluated separately against the submission deadline.

9. Follow-up on Findings

Finding	Source	Result
Sticky search bar suggested	Questionnaire	Implemented: Sticky navbar (#228)
Slow homepage navigation	Clarity session 4	Investigated in the Lighthouse audit; partly addressed (#229 caching, #232 font preconnect). Mobile performance remains below 90
Edit Profile lacks clear feedback on desktop	Clarity session 3	Recommended to the development team; not implemented
Unclear combined filter behavior	Questionnaire	Open; recommended for future work
Delayed filter feedback / results hard to find on mobile	Questionnaire	Open; recommended for future work
Unexpected recipe categorization and unclear tags	Questionnaire	Open; mainly caused by API recipe data
Placeholder or unsuitable recipe content	Questionnaire	Open; content comes from the shared API
Meal Planner and Pantry unfinished	Questionnaire	Pantry implemented (#255); Meal Planner remains future work

10. Conclusion

Usability testing indicates that FlavorCraft Labs provides a workable foundation for recipe discovery and browsing.

Participants successfully explored recipes through multiple navigation paths, and the questionnaire feedback was broadly positive.

The main improvements concern making filtering easier to understand, improving recipe categorization and content quality, and providing clearer feedback for certain interactions.

Combining questionnaires with manually reviewed Clarity recordings provided a more reliable assessment than relying on automated analytics warnings alone.

Final conclusion: The core experience is usable, with identifiable opportunities for refinement. The findings provide actionable guidance for final quality assurance and future development.

Sprint 2 - SEO & Analytics Dependencies & Handoff

Project: FlavorCraft Labs
Role: SEO & Analytics Lead
Sprint: 2 (October 13-19, 2026)

Objective

Implement the planned SEO and analytics functionality, establish a Lighthouse baseline, and begin collecting meaningful behavioral data in time for Sprint 3 analysis and UX improvements.

Several SEO and analytics tasks depend on pages and features developed by other team members. The following overview identifies these dependencies and the coordination needed during Sprint 2.

Required Pages and Features

Page/Feature	**What SEO & Analytics Needs**	**Related Tickets**
Homepage /	A working recipe feed and	Custom events,

	search/filter functionality to test discovery, performance and accessibility	Lighthouse
Recipe detail `/recipes/[id]`	Working, directly accessible dynamic recipe route, API recipe data, and visible recipe content for metadata and JSON-LD implementation.	Dynamic Metadata, Recipe JSON-LD, Lighthouse
Search functionality	A defined search interaction and working navigation from search results to recipe details.	Custom Events
Favorites functionality	Working save/remove functionality and confirmation of successful localStorage updates.	Custom Events
Analytics consent	A working consent solution that exposes whether analytics tracking is permitted and supports withdrawal.	Clarity Integration, Custom Events

Accessible test deployment	Stable pages that can be opened and tested in a browser	Lighthouse, Clarity Testing.

Coordination With Other Team Members

Frontend Developers

- Confirm when the homepage and dynamic recipe detail pages will be available.
- Agree on how recipe data is fetched and made available for SEO metadata and JSON-LD.
- Coordinate where search and Favorites analytics events should be triggered.
- Agree on how the analytics consent state will be exposed to the Clarity integration.
- Notify the SEO & Analytics Lead when relevant features are ready for integration and testing.

Consent Design & Implementation

- Finalize the analytics consent approach, including whether to use a Consent Management Platform.
- Confirm who is responsible for configuring and testing the CMP, if selected.
- Ensure visitors can accept, reject and later withdraw analytics consent.
- Coordinate the consent implementation with the Clarity integration.

Scrum Master / Product Owner

- Confirm that SEO and analytics dependencies are accounted for in Sprint 2 planning.
- Clarify the consent approach and any outstanding client requirements
- Coordinate a realistic testing milestone so behavioral data can be collected before Sprint 3 analysis begin.

Proposed timeline for discussion

Target	**Deliverable**
Early Sprint 2	Confirm route ownership, API integration, consent approach and feature delivery order.
Mid-Sprint 2	Working homepage and recipe detail pages available for SEO implementation and initial Lighthouse testing
Mid-to-late Sprint 2	Consent-aware Clarity integration and available custom analytics events tested
By the end of Sprint 2, ideally earlier	Stable test deployment available for invited testers, with consent-aware

	analytics enabled.
Early Sprint 3	Review recordings, heatmaps and event activity: share prioritized UX recommendations.

These are proposed milestones for discussion, not fixed deadlines.

Outstanding Questions for Sprint Planning

1. Who owns the recipe detail page and when is it expected to be functional?
2. When will search and Favorites be ready for analytics event integration?
3. Will consent be managed through a CMP or a custom implementation, and who owns the technical setup?
4. When will an accessible, stable deployment be available for Lighthouse audits and user testing?
5. Who will recruit and coordinate test users, and when should testing begin?

Current Blockers and Risks

Lighthouse baseline: Could not be completed in Sprint 1 because no testable pages were available. The audit will be carried forward to Sprint 2 once a suitable deployment is available.

Analytics consent: The final implementation approach is pending team/PO confirmation.

Limited analysis window: If testable functionality is delivered late in Sprint 2, there may be insufficient time to collect and analyze meaningful behavioral data before submission.

Feature dependencies: Dynamic SEO requires working recipe detail pages; custom events require working search and Favorites functionality.

Sprint 2 SEO & Analytics Tickets

1. Implement Dynamic Metadata & Open Graph
2. Implement Recipe JSON-LD Structured Data.
3. Integrate Microsoft Clarity Analytics
4. Implement Custom Analytics Events
5. Establish Lighthouse Baseline & SEO QA

Suggested ownership: SEO & Analytics Lead, with collaboration from the relevant feature devs and QA

Sprint 3 follow-up: Analyze collected behavioral data, recommend UX improvements, and repeat Lighthouse audits to evaluate progress toward the client's targets.

Ticket scope, ownership and priority is subject to Sprint Planning and team capacity.

Sprint 2–3 Follow-up — 21 October 2026

The original dependencies and risks above were documented for Sprint 2 planning. The following update reflects subsequent implementation and testing progress.

- **Recipe JSON-LD:** Implemented, merged, and validated using Google's Rich Results Test.
- **Dynamic metadata and Open Graph:** Implemented and locally tested. Verified in production on 22 October.
- **Microsoft Clarity:** Integrated with analytics consent controls. Consent behavior was manually verified.
- **Custom analytics events:** Four events implemented and confirmed in the Clarity dashboard.
- **Lighthouse:** Baseline audits completed, followed by additional performance investigations.
- **User testing:** Seven questionnaire responses collected and six verified Clarity recordings reviewed.

The main initial dependencies were sufficiently resolved to allow SEO implementation, analytics integration, and usability testing.

Detailed results are documented in the SEO, analytics, Lighthouse, and usability testing sections of this document.

SEO & Analytics — Final Handover

Project: FlavorCraft Labs
Role: SEO & Analytics Lead
Status date: 22 October 2026

1. Overview

SEO and analytics work focused on improving recipe discoverability, implementing behavioral analytics, evaluating technical quality, and identifying usability improvements.

This handover summarizes completed work, verification results, outstanding items, and recommendations for further development.

2. SEO Implementation

Recipe JSON-LD Structured Data

- Implemented and merged Recipe structured data.
- Validated two recipe pages using Google's Rich Results Test.
- Both pages returned one valid Recipe item, with optional property warnings.

Status: Implemented and verified.

Dynamic Metadata & Open Graph

- Implemented recipe-specific metadata and Open Graph information.
- Tested locally and verified in production (22 October)

Status: Implemented and verified in production.

3. Analytics Implementation

Microsoft Clarity was integrated with the application's analytics consent mechanism.

Four custom events were implemented:

- `recipe_searched`
- `recipe_opened_from_search`
- `recipe_favorited`
- `recipe_unfavorited`

All four events were confirmed in the Clarity dashboard.

Manual testing verified that no new Clarity tracking requests were observed after rejecting or withdrawing consent, while tracking requests were observed after acceptance.

The Privacy Policy was reviewed for analytics-related disclosures.

Status: Implementation and documented testing completed.

4. Lighthouse & Technical QA

Lighthouse audits were performed for the homepage and authentication pages, followed by additional homepage performance investigations.

The investigation identified areas requiring further review, including server response time, external font loading, recipe image delivery, and navigation performance.

These findings were documented for the development team. Final audit, 22 October 2026 (develop deployment, homepage): Mobile 86.3 average (84 / 86 / 89), Accessibility 97, Best Practices 96, SEO 100. Desktop 98/ 97 / 100 / 100. Resolved: category 404 errors, third-party cookies and DevTools issues. Still open: server response time, render-blocking Adobe Fonts, oversized external images, and both hero images being downloaded.

The 90+ target was met on desktop but not for mobile Performance.

Details in `src/app/docs/seo/lighthouse-audit.md`

Status: Auditing and investigation completed; optimization follow-up recommended.

5. Usability Testing

Usability evaluation included:

- Seven questionnaire responses.
- Six verified Microsoft Clarity tester recordings.

Feedback was generally positive regarding recipe discovery and browsing.

The main improvement opportunities involved combined filtering, recipe categorization, content accuracy, interaction feedback, and homepage performance.

Several Clarity dead-click warnings were manually reviewed and did not represent confirmed defects.

Status: Testing and findings report completed.

6. Outstanding Items

1. Review and prioritize filtering and recipe categorization issues.
2. Evaluate improvements to desktop Edit Profile feedback.
3. Mobile Performance: hero image priority, font loading, external image optimization
4. SEO improvements for future work: sitemap.xml, robots.txt, and canonical URLs.

7. Documentation & Handover

Supporting documentation includes the SEO implementation plan, analytics research and testing records, Lighthouse audit, usability testing instructions, and usability findings report.

The detailed Lighthouse investigation is also documented in `src/app/docs/seo/lighthouse-audit.md`.

This handover reflects the verified project status as of 22 October 2026. Any subsequent merges, fixes, or testing results should be incorporated before final submission.

8. Conclusion

The primary SEO and analytics implementation work has been completed, with structured data validated, analytics events confirmed, and usability testing documented.

The remaining priorities are mobile performance, targeted usability improvements, and the SEO improvements listed above.

The project now has documented findings and recommendations that can support final quality assurance and future development.